

Descriptive Statistics Project

Name: Austin [REDACTED]

Date: 9/18/2025

Instructor: Professor Samuel Chukwuemeka

Project:

Descriptive Statistics of the number of times Major League players were elected/selected to the All-Star game who eventually made the Hall of Fame.

Direct Webpage:

<https://www.statcrunch.com/app/index.html?dataid=3254356>

The names and the number of times they were elected/selected for All Star Games are:

Last Name	First Name	Number of Times Elected/Selected for All Star Games
Aaron	Hank	25
Alomar	Roberto	12
Appling	Luke	7
Bagwell	Jeff	4
Aparicio	Luis	13
Banks	Ernie	14
Bench	Johnny	14

Berra	Yogi	18
Ashburn	Richie	6
Averill	Earl	6

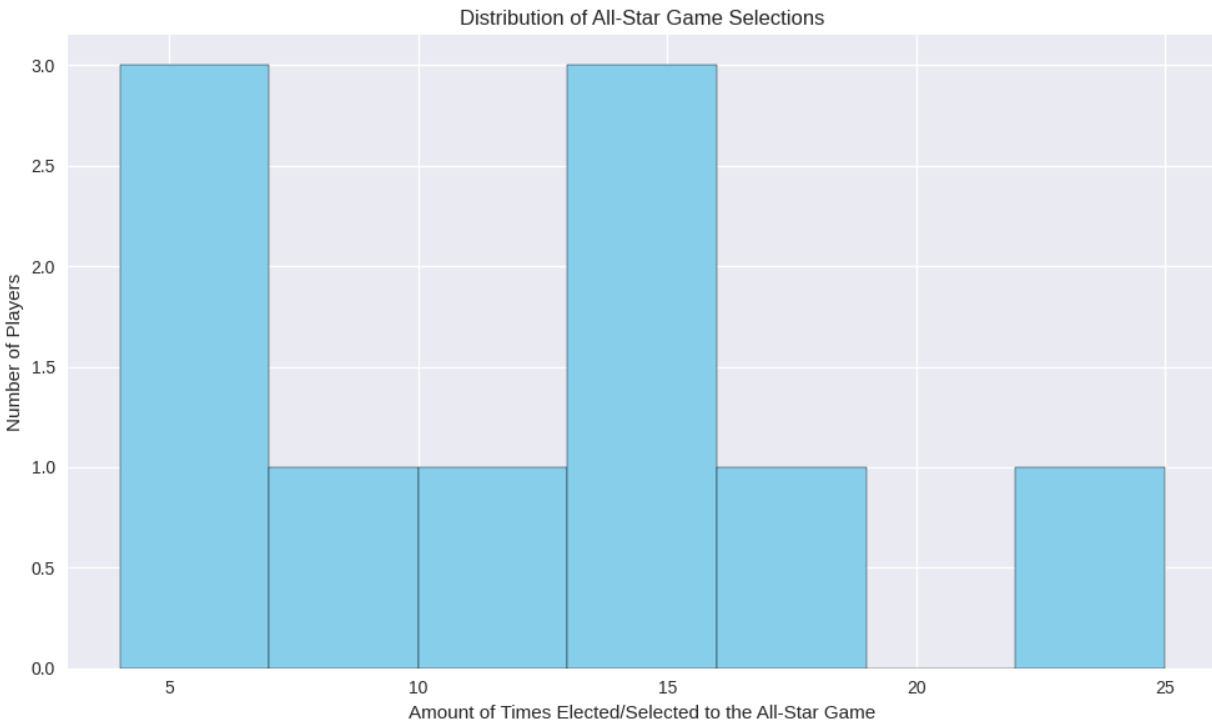
Objectives:

- (1.) Data Presentation: I will present my data using dot plot and describe it.
- (2.) Measures of Center: I will calculate and interpret, the mean, median, mode and midrange of the dataset.
- (3.) Measures of Spread: I will calculate and interpret the range, variance, standard deviation, and interquartile range of the dataset.
- (4.) Measures of Position: I will calculate and interpret the Five-Number Summary of the dataset.

Description: Amount of time Major League Players were elected/selected to the All-Star Game who eventually made the Hall of Fame

Data Values: 25, 12, 7, 4, 13, 14, 14, 18, 6, 6

Presentation:



- (1.) The horizontal axis is the number of times elected/selected to the All-Star Game and the vertical axis is the number of Players elected/selected to that amount of All-Star Games
- (2.) The histogram shows that the most frequent number of times elected/selected to the All-Star Game is between 4 – 6 and 13 – 15
- (3.) There are no outliers in this dataset.

Analysis:

Options												
Summary statistics:												
Column	n	Mean	Median	Mode	Min	Max	Range	Variance	Std. dev.	Q1	Q3	IQR
Number of Times Elected/Selected for All Star Games	10	11.9	12.5	Multiple modes	4	25	21	41.655556	6.4541115	6	14	8

- a) Let x be the number of times Major League Players were elected/selected to the All-Star game

Measures of Center:

$$\text{Mean: } \Sigma x = 25 + 12 + 7 + 4 + 13 + 14 + 14 + 18 + 6 + 6 = 119$$

$$= \frac{119}{10} = 11.9$$

The Mean average amount of times elected/selected to the All-Star Game is 11.9

Median:

I will sort the data in ascending order

4, 6, 6, 7, 12, 13, 14, 14, 18, 25

$$\text{Median} = Q_2 = \frac{12 + 13}{2} = 12.5$$

When the number of times elected/selected to the All-Star Game values are arranged in ascending order, the median value is 12.5 number of times elected/selected to the All-Star Game

Mode: In this dataset the number of times that occur the most frequently is 6 and 14.

Midrange: The minimum number of times elected/selected is 4, the maximum number of times elected/selected is 25

$$\text{Midrange} = \frac{\text{minimum} + \text{maximum}}{2} = \frac{4 + 25}{2} = 14.5$$

The average amount of times elected/selected to the All-Star games of the minimum and maximum is 14.5.

Measures of Spread:

$$\text{Range} = \text{Maximum} - \text{Minimum}$$

$$= 25 - 4$$

$$= 21$$

The difference between the maximum and minimum number of times elected/selected to the All-Star Game is 25.

Variance:

x	$x - \bar{x}$	$(x - \bar{x})^2$
4	-7.9	62.41
6	-5.9	34.81
6	-5.9	34.81
7	-4.9	24.01
12	0.1	0.01
13	1.1	1.21
14	2.1	4.41
14	2.1	4.41
18	6.1	37.21
25	13.1	171.61

$$\Sigma (x - \bar{x})^2 = 374.9$$

$$\text{Variance, } s^2 = \frac{\Sigma(x - \bar{x})^2}{n - 1} = \frac{374.9}{10 - 1} = 41.655556$$

Standard Deviation:

To find the standard deviation we will need to take the square root of the variance

$$\text{Standard Deviation}, s = \sqrt{\text{Variance}} = \sqrt{41.655556} = 6.4541115$$

The value of the standard deviation shows that there is a considerable amount of spread among the number of times players were elected/selected to the All-Star Game.

Interquartile Range:

The interquartile range is the distance between upper and lower quartiles

Data in Ascending Order:

4, 6, 6, 7, 12, 13, 14, 14, 18, 25

$$n = 10$$

$$\frac{1}{4} * 10 = 2.5 \text{ Round up to } 3$$

$$3\text{rd Value} = 6$$

$$\text{Lower Quartile}, Q_1 = 3\text{rd value} = 6$$

$$\frac{3}{4} * 10 = 7.5$$

Round up to 8

$$8th\ value = 14$$

$$Upper\ Quartile, Q_3 = 8th\ value = 14$$

$$Interquartile\ Range, IQR = Q_3 - Q_1$$

$$IQR = 14 - 6 = 8$$

Interquartile Range: The variation in 50% of the number of times the players were elected/selected for the All-Star Game is 8.

Measures of Position:

The 5 number summary of the dataset and their interpretations are:

$$Minimum = 4$$

The minimum number of times players who made the hall of fame were elected/selected to All Star Games is 4

$$Lower\ Quartile, Q_1 = 6$$

4 times is the 25th percentile of the number of times players who made the hall of fame were elected/selected to the All-Star Game.

$$\text{Middle Quartile} = \text{Median}, Q_2 = 12.5$$

12.5 times is the 50th percentile of the number of times players who made the hall of fame were elected/selected to the All-Star Game.

$$\text{Upper Quartile}, Q_3 = 8$$

8 times is the 50th percentile of the number of times players who made the hall of fame were elected/selected to the All-Star Game.

$$\text{Maximum} = 25$$

25 times is the maximum value of the number of times players who made the hall of fame were elected/selected to the All-Star Game

Works Cited (MLA 9)

- Chukwuemeka, S. (n.d.). *Descriptive Statistics* [Review of *Descriptive Statistics*].
Descriptive Statistics Projects. Retrieved November 3, 2025, from
<https://conferencepresentations.appspot.com/Projects/Statistics/DescriptiveStatistics.html>
- “StatCrunch.” *Statcrunch.com*, 2025,
www.statcrunch.com/app/index.html?dataid=3254356.